

NYMEX PALLADIUM FUTURES (PA)

Spread & Curve Structure -- Fundamental Reference

Gasoline Catalysis, Russian Supply Concentration, and the EV Demand
Question

This document presents a fact-based reference on the fundamental drivers of Palladium futures (ticker: PA), contract size 100 troy ounces, traded on NYMEX (a CME Group exchange) alongside platinum, with which it shares the substitution relationship described in detail in the Platinum reference document in this series. Palladium occupies a position that mirrors platinum's in many structural respects -- a platinum group metal (PGM) whose demand is dominated by a single automotive catalytic converter application and whose supply is geographically concentrated in a small number of countries -- while differing from platinum in two specific and consequential ways: palladium's catalyst application is gasoline-engine three-way catalysts rather than diesel oxidation catalysts, putting it on the opposite side of the structural demand trends reshaping the global vehicle fleet, and its supply concentration centres on Russia (via Norilsk Nickel) to a degree that has made it one of the most directly sanctions-exposed major metals markets covered in this document series. This document presents the palladium-specific fundamentals in full, with particular emphasis on Russian supply concentration and sanctions exposure, the gasoline-catalyst demand structure and its exposure to electric vehicle adoption, the historic platinum-palladium price inversion from palladium's side of that relationship, and the documented multi-year deficit and subsequent price collapse that have characterised this market over recent cycles.

Futures First -- Internal Education Material
For Analyst Group Use

1. Contract Specifications

Palladium futures trade on the NYMEX division of CME Group, the same division on which platinum trades (described in the companion Platinum reference document), distinguishing both from gold, silver, and copper, which trade on the COMEX division.

Parameter	Detail
Exchange	NYMEX (New York Mercantile Exchange) -- a CME Group division, the same division on which platinum trades
Ticker	PA (front month)
Contract size	100 troy ounces
Quotation	US dollars per troy ounce (USD/oz); minimum tick = \$0.05/oz = \$5.00 per contract
Contract months	March, June, September, December as the primary actively quoted months (a quarterly cycle similar in structure to platinum's), with additional nearby months also listed.
Settlement type	Physical delivery of palladium (99.95% minimum purity, NYMEX-approved brands/refiners) at NYMEX-approved depositories in the New York area. As with the other precious and platinum group metals covered in this series, the substantial majority of contracts are cash-settled or rolled rather than taken to physical delivery.
Trading hours	CME Globex: Sunday-Friday 18:00-17:00 ET (23-hour session); liquid hours approximately 08:20-13:05 ET
Last trading day	Third-to-last business day of the delivery month
Contract value at \$1,000/oz	\$100,000 per contract
Related contract	Platinum futures (PL), also traded on NYMEX, 50 troy ounces -- palladium's primary substitution partner, described in full in the Platinum reference document, with the platinum-palladium ratio addressed from palladium's perspective in Section 6 of this document.
LPPM relationship	As with platinum, the London Platinum and Palladium Market (LPPM) administers the twice-daily LBMA Palladium Price fix via ICE Benchmark Administration, the OTC physical-market reference alongside the NYMEX futures contract.

1.1 Contract Size Difference From Platinum

- Palladium's 100 troy ounce contract size is double platinum's 50 troy ounce size, a specification difference that traders constructing a platinum-palladium ratio position or spread must account for explicitly when sizing the two legs to achieve a desired notional or risk-equivalent exposure, since a 1:1 contract ratio between PL and PA does not represent an equal-notional position at most price levels.

2. Supply Concentration -- Russia and South Africa as the Two Pillars

Palladium's supply geography substantially mirrors platinum's (described in the companion Platinum reference document), drawn predominantly from the same South African Bushveld Complex operations and Russian Norilsk-Talnakh nickel mining district, but with the relative weighting between the two sources shifted meaningfully toward Russia compared to platinum's heavier South African weighting.

Country	Approx. Share of World Mine Production	Key Structural Facts	PA Curve Relevance
Russia	~40% of world production	Palladium is recovered substantially as a co-product of nickel mining at Norilsk Nickel's Norilsk-Talnakh operations in Siberia -- the world's largest single source of mined palladium -- meaning Russian palladium supply is tied to nickel mining investment and operating decisions rather than to palladium-primary economics, the same by-product supply-inelasticity dynamic described for silver (lead-zinc-copper by-product) in the companion COMEX Silver reference document	The single most important and most geopolitically exposed supply variable in this market; sanctions-related risk (Section 3) is more acute for palladium than for almost any other metal covered in this series
South Africa	~35-40% of world production	Recovered alongside platinum from the same Bushveld Complex operations described in the companion Platinum reference document -- meaning South African labour relations and Eskom electricity supply risk (both described in detail in that document) apply to palladium supply in essentially the same way they apply to platinum, since both metals are co-produced from the same ore body	A secondary, but still very substantial, supply source carrying the same labour and electricity-supply risk factors as platinum
North America (Canada, US)	Remainder, single-digit-to-low-double-digit percentage	Recovered as a co-product of nickel-copper mining, notably in Ontario's Sudbury Basin, mirroring the same by-product relationship described for platinum in the companion reference document	A smaller but non-trivial secondary source

2.1 Why Russia's Share Is Larger for Palladium Than for Platinum

- The Bushveld Complex ore body (South Africa's dominant PGM source, described in the Platinum reference document) is relatively richer in platinum than in palladium, while the Norilsk-Talnakh nickel sulphide ore body (Russia's dominant source) is relatively richer in palladium than in platinum -- a geological fact, rooted in the specific mineralogy of each ore body, that explains why Russia's share of world supply is meaningfully larger for palladium (~40%) than for platinum (~10-15%, per the Platinum reference document), even though both metals are mined substantially as co-products from operations also producing the other.

3. Sanctions and Geopolitical Risk -- The Defining Risk Factor

Because of Russia's outsized ~40% share of world palladium supply (Section 2), and because that supply is dominated by a single major producer (Norilsk Nickel), palladium carries a degree of single-company, single-country geopolitical supply risk with few parallels among the other metals covered in this document series.

3.1 Historical Sanctions Exposure

- **Why palladium has not been directly embargoed despite broader Russian sanctions:** notwithstanding the wide range of sanctions measures imposed on Russia in response to its invasion of Ukraine (including, as described in the Gasoil reference document, the EU's ban on Russian refined petroleum product imports), Russian palladium exports have not been subject to an equivalent direct import ban in major Western markets, reflecting -- per industry and policy commentary -- the absence of a readily available alternative supply source at the scale required to replace Russian volumes without severely disrupting the global automotive catalyst supply chain that depends on palladium, a "too critical to embargo directly" dynamic distinct from the refined-products case described for gasoil, where alternative Middle Eastern and Indian supply was judged sufficient to absorb a full import ban.
- **The market's persistent geopolitical risk premium:** even without a direct embargo, the market has, at various points, priced in a risk premium reflecting the possibility of future sanctions escalation, voluntary self-sanctioning by Western buyers and trading houses (some commercial counterparties choosing to reduce Russian-sourced exposure even absent a legal requirement to do so), or operational disruption to Russian export logistics -- a continuously monitored geopolitical overlay on top of the standard supply/demand fundamentals described elsewhere in this document, and one that can shift rapidly with news flow in a way that has no equivalent for the more geographically diversified metals covered elsewhere in this series.

3.2 Norilsk Nickel as a Single-Company Risk Concentration

- Because Norilsk Nickel is the single dominant producer within the dominant producing country, company-specific operational news (smelter incidents, environmental disclosures, production guidance changes, and the company's own corporate and ownership developments) carries a degree of market-moving significance for palladium with few parallels among the more fragmented producer bases described for other metals in this series, where no single company holds a comparably dominant share of world supply.

4. Demand Fundamentals -- Gasoline Catalysis as the Dominant Application

As described in the Platinum reference document's discussion of catalyst chemistry, palladium is the preferred catalyst metal for gasoline-engine three-way catalytic converters, putting palladium demand on the opposite side of the global diesel-versus-gasoline vehicle mix shift from platinum.

Application	Approx. Share of Total Palladium Demand	Key Driver	PA Curve Sensitivity
Automotive catalytic converters (gasoline)	~80-85%, a substantially higher concentration than platinum's automotive share	Global gasoline-engine vehicle production volume; tightening global emissions standards, which have historically required higher PGM loading per vehicle to meet stricter limits	By far the dominant demand category -- palladium's automotive demand concentration is meaningfully higher than platinum's, making global gasoline vehicle production data an even more singularly important demand signal for PA than the equivalent automotive data is for PL
Electronics	~5-8%	Palladium is used in multilayer ceramic capacitors (MLCCs) and other electronic components, tying this demand component to global consumer electronics and semiconductor-adjacent production cycles	A secondary demand component, more cyclically tied to electronics/semiconductor production trends than to the automotive cycle
Dental and other industrial/medical	~3-5%	Dental alloy applications and other smaller industrial uses	A minor, slow-moving demand component
Jewellery	~3-5%, smaller than platinum's jewellery share	A smaller jewellery demand base than platinum, reflecting palladium's historically lower profile as a standalone jewellery metal	A minor demand component relative to palladium's overwhelming automotive weighting
Investment	Small, variable	Palladium-backed ETF and physical investment demand, smaller in scale than the equivalent platinum and gold investment markets	A minor swing factor, generally less significant than for platinum or gold

4.1 Tightening Emissions Standards as a Historical Demand Tailwind

- Progressively tightening global vehicle emissions standards (across major markets including the EU, US, and China) have, over an extended historical period, required increased PGM loading per catalytic converter to meet stricter emissions limits -- a regulatory-driven demand tailwind that operated alongside, and at times outweighed, underlying vehicle production volume growth itself, meaning palladium demand growth has historically outpaced gasoline vehicle production growth alone during periods of standards tightening.

4.2 Electric Vehicle Adoption as the Dominant Long-Run Demand Risk

- Because battery electric vehicles require no catalytic converter at all, and because palladium's automotive demand concentration is even higher than platinum's (Section 4 table), the global shift

toward EV adoption represents a larger proportional long-run demand risk for palladium than for platinum -- a structural consideration that has, at various points, weighed on long-run palladium demand expectations and contributed to the price dynamics described in Section 6.

- **The absence of an offsetting hydrogen fuel cell demand channel:** unlike platinum, which has a documented, if currently small-scale, emerging demand channel in hydrogen fuel cell catalysis (described in the Platinum reference document), palladium has no equivalent offsetting structural demand growth narrative -- meaning the EV-adoption headwind described above is not balanced by a comparable structural tailwind in the way platinum's diesel-decline headwind is partially offset by its fuel-cell-demand narrative.

5. The Forward Curve and Carry Structure

Palladium's forward curve reflects a carry framework broadly similar to platinum's, with the lease/borrow market for palladium specifically generally regarded as thinner and less liquid than for platinum, gold, or silver, given palladium's smaller overall market size and more concentrated holder base.

5.1 Backwardation and the Multi-Year Deficit Narrative

- **The historical structural deficit narrative:** palladium experienced a documented, multi-year period of structural market deficits (demand exceeding mine and recycled supply) during the years of peak emissions-standards-driven demand growth described in Section 4.1, drawing down above-ground inventories and contributing to a sustained period of elevated prices and, at various points, backwardation in the futures curve -- a deficit narrative structurally analogous in framing (though arising from different underlying drivers) to the silver and copper structural deficit narratives described in the companion COMEX reference documents.
- **The subsequent price normalisation:** as the EV-adoption demand headwind described in Section 4.2 became a more dominant market narrative, and as the deficit period's extreme pricing itself encouraged demand substitution and thrifting efforts by catalyst manufacturers (reducing PGM loading per converter where technically feasible, the same general thrifting dynamic described for silver in solar panel manufacturing in the companion COMEX Silver reference document), palladium prices subsequently underwent a sustained decline from their peak levels -- the specific magnitude and timing of which should be verified against current data given this reference document's knowledge cutoff, but the underlying demand-substitution and thrifting mechanism described here remains structurally relevant regardless.

5.2 Recycling as a Significant Secondary Supply Source

- Because spent automotive catalytic converters contain recoverable PGM content, secondary supply from end-of-life vehicle catalyst recycling is a meaningful and closely tracked component of the total palladium supply picture, with recycling volumes responding, with a lag, to the size of the historical vehicle fleet now reaching scrappage age -- a supply source with its own multi-year dynamics distinct from primary mine supply, and one that has, per industry analysis, grown in relative importance to the overall supply balance.

6. The Platinum-Palladium Relationship From Palladium's Side

The platinum-palladium substitution relationship is described in full, including the underlying catalyst chemistry, in the companion Platinum reference document; this section summarises the relationship specifically from palladium's perspective.

6.1 The Historical Premium Inversion

- As described in the Platinum reference document, palladium traded at a sustained discount to platinum for most of their shared trading history, before a multi-year period in which palladium traded at a premium to platinum -- a direct consequence of the demand-growth divergence described in Section 4.1 (palladium's gasoline-catalyst demand benefiting from tightening emissions standards and a growing gasoline-vehicle production base, even as platinum's diesel-catalyst demand faced the structural decline described in the Platinum reference document's discussion of the European diesel share decline).
- **The relationship's subsequent evolution:** as the EV-adoption headwind described in Section 4.2 became more pronounced for palladium specifically, and as the deficit-driven price level itself encouraged the substitution and thrifting dynamics described in Section 5.1, the platinum-palladium relationship has continued to evolve -- the specific current relationship should always be verified against live pricing rather than assumed from any historical description, given how substantially this relationship has moved over the past decade.

6.2 Substitution Flexibility -- Loading Ratio Adjustments Within Engine Type

- As described in the Platinum reference document, full platinum-palladium substitution across the diesel/gasoline divide is constrained by catalyst chemistry rather than being a pure cost-driven choice; however, catalyst formulators retain some flexibility to adjust the specific platinum-palladium-rhodium loading mix within a given engine-type application in response to relative metal prices, meaning a sustained, extreme palladium premium over platinum has, at various points, incentivised catalyst manufacturers to substitute a greater share of platinum into gasoline-engine catalyst formulations where the chemistry allows it -- a margin-level substitution response that operates as a partial demand-side moderating mechanism on extreme palladium pricing, though one that operates with a meaningful lag given vehicle platform development and certification timelines.

7. Cross-Market Relationships

7.1 Rhodium as a Related, Smaller Market

- Rhodium -- another platinum group metal, used alongside platinum and palladium in automotive catalysts, particularly for nitrogen oxide (NOx) reduction -- is a much smaller, thinner market without an actively liquid, widely-traded futures contract comparable to platinum or palladium's NYMEX listings, but is tracked by the same commercial participants (catalyst formulators, PGM trading houses) given its role in the same automotive catalyst application, making rhodium price trends a relevant qualitative cross-check on broader PGM demand conditions even without an equivalent liquid futures market for direct cross-market spread construction.

7.2 Russian Ruble (RUB/USD) and Sanctions-Adjacent Currency Risk

- As with other Russian-sourced commodities, RUB/USD movement and the broader currency and capital-control environment affecting Russian exporters is a relevant background variable for Norilsk Nickel's palladium export economics, though this channel should be considered alongside, and generally as secondary to, the direct sanctions and geopolitical risk factors described in Section 3, which are more often the dominant driver of Russian-supply-related palladium price action than pure currency-cost mechanics.

7.3 Global Automotive Production and Semiconductor Supply

- Because automotive demand dominates palladium consumption (Section 4), broader automotive production disruptions -- including the well-documented global semiconductor shortage affecting vehicle production volumes during the early 2020s -- are a relevant demand-side cross-market consideration, operating through vehicle production volume rather than through any direct commodity-market linkage.

7.4 Speculative Positioning

- **CFTC COT for NYMEX Palladium:** weekly managed money net positioning is tracked, consistent with the positioning framework described throughout this document series. Palladium's smaller market size and thinner liquidity relative to gold and even platinum (Section 5) means positioning extremes have, at times, been associated with sharper and more volatile price moves than equivalent positioning levels in larger precious metals markets.

8. Documented Historical Events

- **The multi-year structural deficit period:** as described in Section 5.1, a sustained period of demand exceeding mine and recycled supply, driven by tightening global emissions standards, produced a multi-year drawdown in above-ground inventories and a sustained period of elevated prices and the historic premium over platinum.
- **The 2022 Russian invasion of Ukraine and subsequent sanctions environment:** as described in Section 3.1, the broader sanctions response to the invasion created sustained market attention to Russian palladium supply risk, even though palladium itself was not subject to a direct Western import ban, given the absence of a readily available alternative supply source at the required scale.
- **The subsequent price normalisation from peak levels:** as described in Section 5.1, the combination of EV-adoption demand concerns and demand-side thrifting/substitution responses to the historically elevated price level contributed to a sustained decline from palladium's peak pricing -- the specific magnitude and current price level should be verified against live data rather than assumed from this reference document.

9. Key Data Sources and Release Schedule

Report / Data	Publisher	Frequency	Release Timing	Primary PA Curve Impact
Norilsk Nickel production guidance and operational updates	Norilsk Nickel (company disclosures)	Quarterly + as announced	Variable	All contracts -- the single most important supply-side disclosure given Norilsk's dominant production share
Sanctions and trade policy announcements (Russia-related)	US Treasury OFAC; EU Commission; UK government	As scheduled/ amended	Variable	All contracts -- the dominant geopolitical risk variable described in Section 3
World Platinum Investment Council (WPIC) quarterly reports	WPIC	Quarterly	Variable	All contracts -- covers palladium alongside platinum in its broader PGM market balance reporting
Global light vehicle production data (gasoline-engine specifically)	Industry associations; company disclosures	Monthly/periodic	Variable	All contracts -- the dominant demand-side data series given palladium's automotive concentration
Global EV adoption and sales data	Industry associations; national vehicle registration data	Monthly/periodic	Variable	Back curve -- the structural long-run demand risk variable described in Section 4.2
Emissions standards regulatory announcements (EU, US, China)	Relevant national/regional regulators	As scheduled/ amended	Variable	Mid-to-back contracts -- PGM loading requirement trajectory
NYMEX registered/eligible palladium stocks	CME Group	Daily	After close	Front contract -- delivery and physical tightness signal
LBMA/LPPM Palladium Price (AM/PM fix)	IBA via LBMA/LPPM	Twice daily	~09:45 / 14:00 London	Spot reference -- physical market benchmark
CFTC Commitments of Traders (Palladium)	CFTC	Weekly	Fri ~15:30 ET (3-day lag)	Positioning / crowding -- given palladium's thin liquidity, extremes can be particularly impactful

NYMEX Palladium shares platinum's industrial-commodity analytical framework and single-application automotive demand concentration, but with two consequential differences: its gasoline-catalyst application places it on the opposite side of the diesel-versus-gasoline structural demand split from platinum, and its supply concentration in Russia -- via the single dominant producer Norilsk Nickel -- gives it a degree of geopolitical and sanctions-adjacent risk exposure with few parallels among the other metals covered in this document series. The historic platinum-palladium premium inversion, and palladium's subsequent price normalisation as EV adoption concerns and demand-side thrifting responded to a multi-year structural deficit, together constitute one of the most significant documented repricing episodes among the metals markets covered in this series.

For spread traders, the most reliable framework treats palladium as an even more concentrated automotive-demand and single-producer-supply-risk market than platinum, with Norilsk Nickel operational disclosures and the broader Russia sanctions environment as the two variables warranting the most continuous attention, alongside global gasoline-vehicle production trends and the EV-adoption trajectory as the dominant long-run demand consideration. The platinum-palladium ratio remains the primary cross-market relationship, though its current level and recent trajectory should always be

checked against live data rather than assumed from any historical description, given the scale of the repricing this relationship has undergone over the past decade.